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NPS/001/022 Technical Specification for Aeolian Vibration Dampers.

1. Purpose

This specification details the CE-Electric UK requirements for the supply of Aeolian vibration dampers for use on the CE Electric UK Distribution System.

This document supersedes the following documents, all copies of which should be destroyed.

Ref	Version	Title

2. Scope

This Specification covers the design, manufacture, installation and testing of Aeolian Vibration dampers for conductors for use on overhead lines located within CE Electric UK.

3. Technical specification

3.1 General

All Aeolian Vibration dampers shall be so designed so as to:

- a) Damp Aeolian Vibration and avoid damage or corrosion to the conductor or individual strands during installation or during service;
- b) Withstand the mechanical loads during installation, maintenance and specified service conditions including conductor temperature variations, ultraviolet radiation, ozone and atmospheric pollutants applicable to the site;
- c) Be free from unacceptable levels of corona and radio interference under all service conditions;
- d) Avoid audible noise under all weather conditions;
- f) Maintain its function over the entire service temperature range from -20°C to 90°C;
- g) Be capable of being removed and reinstalled without damaging the conductor;
- h) Be suitable for safe and easy installation. The clamp design shall retain all parts when opened for attachment to the conductor and shall be designed to ensure the damper, during installation, can be suspended from the conductor before tightening the clamp;
- i) Ensure individual components will not come loose in service.

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3.1.1 Conductor Range

Appendix B provides details on the type and range of Aeolian Vibration dampers required for us on the company's network.

3.2 Specific requirements

3.2.1 Project details

The specific requirements for Aeolian Vibration dampers will depend on the following factors: the geographical orientation of the line with respect to large bodies of water, the frequency of laminar winds (0.5 m/s to 10 m/s), the ground terrain, the nature of the ground cover and the 'everyday' conductor tension.

The use of conventional one response Stockbridge type Aeolian vibration dampers, i.e. with symmetrical damper weights and a 7-strand messenger cable is normally satisfactory in areas with a ground roughness category IV ref BS 8100-1 or higher and an EDT of approximately 20 % of the conductor's rated strength. However, consideration should be given to the use of multi-response Aeolian vibration dampers, i.e. with asymmetrical damper weights and/or spacing and 19-strand messenger cable in areas with a ground roughness category type III or lower, or where a higher EDT is used.

3.2.2 Aeolian vibration limits

When installed in accordance with the supplier's recommendations the vibration damper(s) shall limit the Aeolian vibration levels so that the dynamic bending strain on the surface of the outer wires of the conductor shall not exceed 300 micro-strains (peak to peak) at the vibration damper clamp and at the adjacent suspension clamp or dead-end tension joint / tension clamp. This requirement shall be met for all frequencies up to and including

$$f = \frac{1480}{d} \text{ Hz (where } d \text{ is the conductor diameter in mm).}$$

The supplier shall provide either suitable laboratory test results, field test results or calculations to demonstrate to the company's satisfaction that this requirement is met for each Aeolian vibration damper - conductor combination.

3.2.3 Conductor clamp

Stockbridge type Aeolian vibration damper clamps shall be designed to ensure that after the bolt has been correctly tightened, no slackening of the clamp can occur in service. The design of the clamp shall take account that all Stockbridge type Aeolian vibration dampers will be installed with grease filling the interfaces between the clamp and the conductor to prevent the ingress of moisture. Wherever possible shear head clamp bolts shall be used.

3.2.4 Messenger cable

The messenger cable for Stockbridge type Aeolian vibration dampers shall comprise stranded fatigue resistant high tensile steel wires. The cable shall be protected against corrosion either by sleeving over the complete length of the cable, or by alternative means that have shown to have proven fatigue and corrosion resistance in the UK or similar environmental areas for a minimum of 20 years service life. Details of the alternative corrosion protection shall be submitted to the company.

3.2.5 Damper weights

The design of the Stockbridge type Aeolian vibration damper shall ensure that no contact occurs between damper

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weights and the messenger cable under service conditions.

3.3 Materials

Materials used in the manufacture of Aeolian Vibration dampers shall have the following properties as appropriate:

- a) Adequate strength for the intended application and service life requirements (including mechanical loads, vibrations, electrical currents and environmental effects) and free from defects that would affect the performance of the Aeolian Vibration damper;
- b) Not be liable to intergranular or stress corrosion;
- c) Compatible with the conductors' material such that there can be no deteriorious effects on the conductor or Aeolian vibration dampers from their use;
- d) Shall not be adversely affected in the long term by a coating applied for the corrosion protection;
- e) Non-metallic materials shall have good resistance to ageing and be capable of withstanding service conditions without a detrimental change in their properties. The materials used shall have adequate resistance to the effects of nitrogen oxides, ozone, ultraviolet radiation and air pollution over the whole range of service temperatures from -20°C to 90°C. They shall not induce corrosion in materials in contact with them.

3.4 Workmanship

Contract drawings for Aeolian vibration dampers shall in addition to the dimensions, show material types and grades, protective treatment and any other pertinent information.

The Aeolian vibration dampers shall be free from defects and irregularities. Aeolian vibration damper weights and conductor clamps shall have all outside surfaces smooth and all edges and corners well rounded.

Contract drawings shall be submitted to the company.

3.5 Corrosion Protection

All materials used in the manufacture of Aeolian vibration dampers shall be inherently resistant to atmospheric corrosion which could affect their performance.

All ferrous materials, shall either be inherently resistant to atmospheric corrosion, or be suitably protected against corrosion that can occur during transportation, storage or service. All ferrous parts which will be exposed to the atmosphere in service, except those made of the appropriate grade of stainless steel, shall be protected by hot dip galvanizing in accordance with the requirements of BS EN ISO 1461.

Where zinc coated steel messenger wires are used they shall be hot dip galvanized in accordance with the requirements of BS EN 50189.

All external threads shall be cut or rolled before hot-dipped galvanizing. Nuts to be galvanized shall be subsequently tapped 0.4 mm oversize and the threads oiled.

Drain holes (minimum 6 mm diameter) shall be provided where applicable to ensure that any water entering the

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damper weights can escape.

3.6 Installation

The supplier shall provide comprehensive instructions in a suitable format covering the selection and installation of the Aeolian vibration damper, including the number of Aeolian vibration dampers per span, distance from the suspension clamp or mouth of dead-end tension joint / tension clamp taking into account the possibility of armour rods being fitted and the clamp bolt installation torque.

3.7 Quality Assurance

3.7.1 General

Type, sample and routine tests shall be undertaken as appropriate on the Aeolian vibration dampers in accordance with the requirements on this Specification and BS EN 61897.

3.7.2 Identification and marking

All Stockbridge type Aeolian vibration dampers shall be marked to ensure a system of traceability. Corona free markings shall be in accordance with the requirements of BS EN 61284. In addition, the range of conductor sizes for which the Aeolian vibration damper is intended to be used, together with type of conductor shall be clearly shown.

Installation torque values shall be stated in newton-metres (N m) and shall be marked adjacent to the appropriate fasteners tightened during installation.

Details of the proposed marking system shall be submitted to the Company.

3.7.3 Type tests

3.7.3.1 General

Three samples of each Aeolian vibration damper for each conductor size shall be subjected to the type tests. One sample of each Aeolian vibration damper, subjected to the visual examination, dimensional, materials and finish and corona type tests, shall be retained by the manufacturer for comparison with the production Aeolian vibration dampers.

3.7.3.2 Visual examination, dimensional, materials and finish

Visual examination, dimensional, materials and finish including that of any coating, e.g. galvanizing type tests shall be undertaken in accordance with the requirements of Sections 7.1, 7.2 and 7.3 of BS EN 61897.

3.7.3.3 Clamp slip

Clamp slip tests on Stockbridge type Aeolian vibration dampers shall be undertaken in accordance with the requirements of Section 7.5 of BS EN 61897, except that the clamp shall be installed on a portion of greased conductor.

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3.7.3.4 Clamp bolt tightening

Clamp bolt tightening tests on Stockbridge type Aeolian vibration dampers shall be undertaken in accordance with the requirements of Section 7.7 of BS EN 61897, except that an initial test to 100 % of the specified installation torque shall be undertaken. After the 100 % load has been applied, the clamp shall be removed from the conductor and the conductor inspected. The clamp shall be then reapplied and the test continued in accordance with the requirements of Section 7.7 of BS EN 61897.

3.7.3.5 Damper weight attachment

Damper weight attachment tests on Stockbridge type Aeolian vibration dampers shall be undertaken in accordance with the requirements of Section 7.8 of BS EN 61897.

3.7.3.6 Clamp attachment

Clamp attachment tests on Stockbridge type Aeolian vibration dampers shall be undertaken in accordance with the requirements of Section 7.9 of BS EN 61897.

3.7.3.7 Corona

Visible corona type tests shall be undertaken on phase conductor Aeolian vibration dampers in accordance with the requirements of Section 14 of BS EN 61284.

The Contractor / Manufacturer shall submit details of his proposed spacing between sub-conductors and the electrical clearance between the Aeolian vibration dampers and the earthed metal work. Clearances to other objects, whether earthed or at high voltage, shall be sufficient to ensure insignificant electrical field effects on the test fitting(s).

3.7.3.8 Damper characteristics

Aeolian vibration damper performance tests shall be undertaken in accordance with variant B of Section 7.11.1 of BS EN 61897. To establish the characteristics of the Aeolian vibration dampers used in the damping effectiveness evaluation and the damping fatigue tests, damper characteristics tests shall be undertaken in accordance with the recommendations of Section 7.11.2 of BS EN 61897.

3.7.3.9 Damping effectiveness evaluation

The determination of Aeolian vibration damper damping effectiveness shall be undertaken by laboratory test. The laboratory test shall be undertaken in accordance with the recommendations of Section 7.11.3.2 of BS EN 61897. Prior to installation on the laboratory test span the test Aeolian vibration damper shall have been previously subjected to the 'Characteristic' test specified in the preceding clause.

3.7.3.10 Damper fatigue test

The Aeolian vibration damper fatigue test may be undertaken either using the swept frequency method or the resonant frequency method. The test shall be undertaken in accordance with the recommendations of Section 7.12.2 or Section 7.12.3 of BS EN 61897.

3.7.4 Sample tests

Visual examination, dimensional, materials and finish including that of any coating, e.g. galvanizing and mechanical sample tests on Aeolian vibration dampers shall be undertaken in accordance with the requirements of this Specification and BS EN 61897.

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Visual examination shall also include comparison with archived type test sample.

Where specified, similar test procedures to those used for type tests shall be repeated for sample tests.

The acceptance criteria for the sample damper effectiveness evaluation test shall be in accordance with the requirements of Table 2 variant B of BS EN 61897.

4. References

4.1 External Documentation

Reference	Title
BS EN ISO 1461	Hot dip galvanized coatings on fabricated iron and steel articles – Specifications and test methods.
BS EN 50189	Conductors for overhead lines – Zinc coated steel wires.
BS EN 61284	Overhead lines – Requirements and tests for fittings.
BS EN 61897	Overhead lines – Requirements and tests for Stockbridge type Aeolian vibration dampers.
BS 8100-1 1986	Lattice towers and masts. Code of practice for loading Code of practice for loading

4.2 Internal documentation

Reference	Title
1091010188	Aeolian Vibration Dampers

5. Definitions

Term	Definition
Stockbridge type Aeolian vibration damper	A device comprising a messenger cable with a weight at each end and one bolted clamp, attached to a conductor for the purpose of damping Aeolian vibration.

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6. Authority for issue

6.1 Author

I sign to confirm that I have completed and checked this document and I am satisfied with its content and submit it for approval and authorisation.

		Sign	Date
G Hammel	Overhead Line Standards Manager	G Hammel	28/09/07

6.2 Policy Sponsor

I sign to confirm that I am satisfied with all aspects of the content and preparation of this document and submit it for approval and authorisation.

		Sign	Date
G Bolden	Purchasing Manager	G Bolden	01/10/07

6.3 Technical Assurance

I sign to confirm that I am satisfied with all aspects of the content and preparation of this document and submit it for approval and authorisation.

		Sign	Date
S Salkeld	Standards and Assessment Engineer	S Salkeld	28/09/07

6.4 DBD Assurance

I sign to confirm that this document has been assured for issue on to the DBD system

		Sign	Date
S Johnson	DBD Administrator	S Johnson	18/10/07

6.5 Approval

Approval is given for the content of this document

		Sign	Date
C Holdsworth	Standards and Assessment Manager	C Holdsworth	01/10/07

6.6 Authorisation

Authorisation is granted for publication of this document

		Sign	Date
C Morgan	Head of Procurement	C Morgan	05/10/07

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APPENDIX A **SELF CERTIFICATION CONFORMANCE DECLARATION**

Aeolian vibration dampers covered by BS EN 61897 shall comply with the latest issues of the relevant international and British Standards.

This check sheet identifies the clauses within those documents. The manufacturer shall declare conformance or otherwise, clause by clause, using the following levels of conformance declaration codes.

Conformance declaration codes

N/A = Clause is not applicable/ appropriate to the product

Cs1 = The product conforms fully with the requirements of this clause

Cs2 = The product conforms partially with the requirements of this clause

Cs3 = The product does not conform to the requirements of this clause

Cs4 = The product does not currently conform to the requirements of this clause, but the manufacturer proposes to modify and test the product in order to conform.

Instructions for completion

- When Cs1 code is entered no remark is necessary
- When any other code is entered the reason for non-conformance shall be entered
- Prefix each remark with the relevant 'BS EN' or 'ENATS' as appropriate

Manufacturer:

Product Reference:

Name:

Signature:

Date:

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Aeolian vibration dampers

Table 1

	BS EN 61897		
Clause / Sub-clause	Type Test Requirements	Conformance Code	Remarks
4.2	Materials		
4.6	Marking		
4.7	Installation Instructions		
7.1	Visual examination		
7.2	Verification of dimensions, materials and mass		
7.3	Corrosion protection tests		
7.4	Non destructive tests		
7.5	Clamp slip test		
7.6	Breakaway bolt tests		
7.7	Clamp bolt tightening test		
7.8	Attachment of weights to messenger cable tests		
7.9	Attachment of clamp to messenger cable test		
7.10	Corona and radio interference tests		
7.11	Damper performance tests - Damper characteristic tests - Damper effectiveness evaluation		
7.12	Damper fatigue tests		

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Appendix B

Aeolian Vibration dampers

